

Identification and Importance of Brown Adipose Tissue in Adult Humans

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BACKGROUND Obesity results from an imbalance between energy intake and expenditure. In rodents and newborn humans, brown adipose tissue helps regulate energy expenditure by thermogenesis mediated by the expression of uncoupling protein 1 (UCP1), but brown adipose tissue has been considered to have no physiologic relevance in adult humans. **METHODS** We analyzed 3640 consecutive 18F-fluorodeoxyglucose (18F-FDG) positron-emission tomographic and computed tomographic (PET-CT) scans performed for various diagnostic reasons in 1972 patients for the presence of substantial depots of putative brown adipose tissue. Such depots were defined as collections of tissue that were more than 4 mm in diameter, had the density of adipose tissue according to CT, and had maximal standardized uptake values of 18F-FDG of at least 2.0 g per milliliter, indicating high metabolic activity. Clinical indexes were recorded and compared with those of date-matched controls. Immunostaining for UCP1 was performed on biopsy specimens from the neck and supraclavicular regions in patients undergoing surgery. **RESULTS** Substantial depots of brown adipose tissue were identified by PET-CT in a region extending from the anterior neck to the thorax. Tissue from this region had UCP1-immunopositive, multilocular adipocytes indicating brown adipose tissue. Positive scans were seen in 76 of 1013 women (7.5%) and 30 of 959 men (3.1%), corresponding to a female:male ratio greater than 2:1 ($P < 0.001$). Women also had a greater mass of brown adipose tissue and higher 18F-FDG uptake activity. The probability of the detection of brown adipose tissue was inversely correlated with years of age ($P < 0.001$), outdoor temperature at the time of the scan ($P = 0.02$), beta-blocker use ($P < 0.001$), and among older patients, body-mass index ($P = 0.007$). **CONCLUSIONS** Defined regions of functionally active brown adipose tissue are present in adult humans, are more frequent in women than in men, and may be quantified noninvasively with the use of 18F-FDG PET-CT. Most important, the amount of brown adipose tissue is inversely correlated with body-mass index, especially in older people, suggesting a potential role of brown adipose tissue in adult human metabolism.